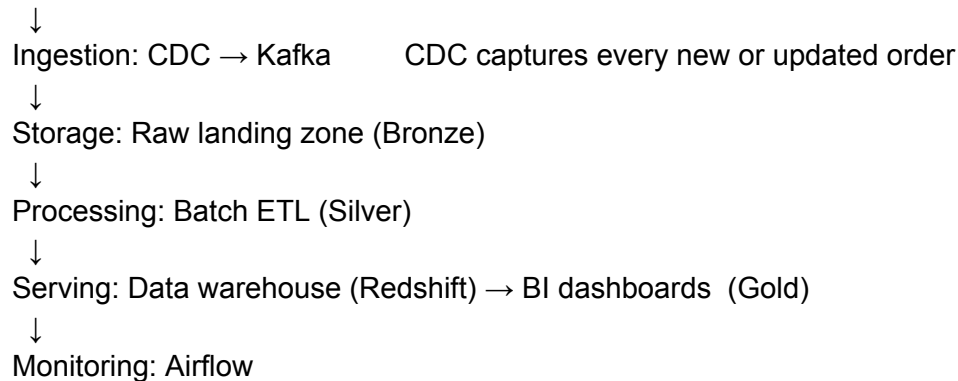


Lab 1: Pipeline Components

An e-commerce website sends order events to a transactional database. Daily product catalog CSV files arrive from vendors. User clicks are logged to an event log collection system.

1. Orders DB

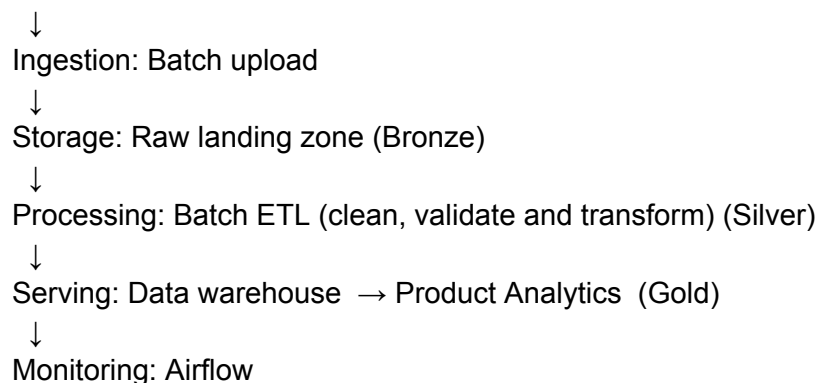
Orders DB (Transactional DB)



- Orders are stored in a transactional (OLTP) database.
- CDC captures every new or updated order without copying the whole table.
- Data is stored in the Bronze layer, cleaned in Silver, and analyzed from the warehouse.

2. Product Catalog CSV Files

Vendor Product CSV Files



- CSV files arrive everyday, so batch ingestion is appropriate.
- Data is cleaned before loading into analytics tables.

3. CClick Logs

User Click Logs



Ingestion: Kafka



Storage: Raw landing zone (Bronze)



Processing: Streaming Spark (Silver)



Serving: Real time dashboard (Gold)



Monitoring: AirFlow

- Clicks are generated continuously, so Apache Kafka is the right choice to collect, store, and process real-time data. Streaming processing provides near real-time metrics.

Lab 2: Classify systems as OLTP or OLAP

Systems:

1. Customer profile database used by customer support for updates.
2. Daily sales reporting database used by analysts for trend analysis.
3. Session store for active web sessions.
4. Historical analytics store for machine learning training.

System	Classification	Reason
Customer profile database used by customer support for updates.	OLTP	handles customer updates, requires frequent and real time transactional data updates.
Daily sales reporting database used by analysts for trend analysis.	OLAP	Used for reporting, data that took place "in the past". Does not require data update. Only reading.
Session store for active web sessions.	OLTP	Stores user sessions, requires low latency and frequent updates.
Historical analytics store for machine learning training.	OLAP	Stores large volume of data for analysis, again data that belongs" to the past"

Lab 3: Classify processing Batch or Streaming

Task	Classification	Reason
Generate daily sales summary for dashboard (end-of-day).	Batch	Pulling a report from data that already happen (not real time)
Detect fraudulent transactions within 5 seconds.	Streaming	It requires real time data to be gathered and low latency.
Aggregate hourly user activity for retention metrics.	Batch	A cost efficient approach would use batch processing because processing can be scheduled instead of running continuously.
Enrich clickstream events with user profile in near-real-time.	Streaming	Click events arrive continuously and require real-time analytics and dashboards.

Lab 4 : ETL vs ELT decision for a use case

Scenario / Input

- Files arrive daily.
- Some fields are missing, column order varies.
- Analysts want to run ad-hoc queries across all products quickly.

Considering the scenario above, I would think the best approach would be ELT because it allows you to upload raw data, store it on a bronze layer and process it later. Otherwise, using ETL, would cost a lot of time to adjust the transformation every time there would be a different column or a missing field.

Lab 5: Map sources to sources → ingestion → storage → processing → serving

1. **Relational orders DB** → **CDC** → **Bronze** → **Streaming** → **Data Warehouse (Snowflake)**
2. **APIs** → **API Pull** → **Bronze (JSON)** → **Batch ETL** → **Data Warehouse (Snowflake)**
3. **Vendor CSV product files** → **Batch upload** → **Bronze** → **Batch ETL** → **Data Warehouse**
4. **Mobile App Events** → **Kafka** → **Bronze** → **Streaming** → **Real Time Dashboard**
5. **Product Catalogue in Dynamo DB** → **Dynamo Streams** → **Bronze** → **Batch ETL** → **Data Warehouse/ Product Analytics**

DynamoDB is a NoSQL database. DynamoDB Streams capture changes as they happen.

Lab 6: Design a batch e-commerce pipeline (orders)

